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Soil Monitoring System

A Real-Time / Field-Based Research Project (22AM284) report submitted to the Jawaharlal Nehru Technological University, Hyderabad

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ABSTRACT

Farming is unpredictable, and soil health plays a crucial role in determining crop success. The Soil Monitoring System is a smart web-based tool designed to help farmers track essential soil parameters like moisture levels, pH balance, temperature, and nutrient concentrations (NPK) in real time. Instead of relying on guesswork or time-consuming manual testing, farmers get instant insights that guide them in making the right decisions for irrigation, fertilization, and crop management.

The system provides real-time alerts when soil conditions change unexpectedly, integrates weather data for better predictions, and features an interactive dashboard to track historical trends. Designed to be easy to use and accessible across devices, it ensures farmers can monitor their fields anytime, anywhere. By reducing resource waste and improving yields, the system promotes sustainable farming and smarter agricultural practices.